Neural Network Classification Method for Remote Sensing Images Based on Data Augmentation and CBAM

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***Abstract*: The classification and recognition of remote sensing images are of great significance for territorial planning, agricultural monitoring, ecological assessment, disaster early warning, and other fields. Currently, several neural network-based methods for remote sensing image classification exist, but their accuracy still has room for improvement. In this paper, based on data augmentation, the Convolutional Block Attention Module (CBAM), Generalized Mean (GeM) pooling, and small-kernel stem are comprehensively used to enhance the model's fitting ability and generalization ability. The proposed model is evaluated on the EuroSAT (European Space Agency Satellite Image Dataset) dataset. Experimental results show that the classification accuracy of the proposed model on the validation set (96.72%) is significantly higher than that of the baseline (94.57%). Additionally, ablation experiments are conducted to verify the effectiveness of data augmentation and model modifications in improving classification accuracy.**

Keywords: Remote sensing images;Neural networks; Classification; Convolutional Block Attention Module (CBAM)

I. Introduction

The classification and recognition of remote sensing images play a critical role in environmental monitoring, urban planning, agriculture, and disaster management. With the advancement of remote sensing technology, automated image classification can provide accurate information support for real-time monitoring and large-scale data processing, thereby optimizing decision-making processes[1]. P. Xu et al. [2] pointed out that accurate, high-resolution Land Cover (LC) information is essential for addressing contemporary environmental challenges; as satellite data acquisition, cloud processing, and deep learning technologies progress, the production of high-resolution Global Land Cover (GLC) maps has become increasingly feasible.

Although recent advancements in Convolutional Neural Networks (CNNs) and Transformers have significantly improved classification accuracy, remote sensing scene classification remains challenging. For example, K. Chen et al. [4] proposed RSMamba (Remote Sensing Mamba), a novel architecture for remote sensing image classification that exhibits excellent performance on multiple datasets and lays a foundation for future visual foundation models. M. Mehmood [3] noted that remote sensing is widely used for surveying dam, bridge, and pipeline sites, locating construction materials, and providing detailed geographical information, while also summarizing current trends and future research directions in remote sensing image classification.

In recent years, neural network-based remote sensing image classification methods have advanced, but accuracy gaps persist. Z. Shao et al. [5] (2018) highlighted that Earth observation satellites capture remote sensing images with different spatial and spectral resolutions (e.g., panchromatic (PAN) images with high spatial resolution and multispectral (MS) images with high spectral resolution). M. Kumari [6] (2023) emphasized that while deep learning has achieved remarkable progress in artificial intelligence fields (e.g., image recognition, speech recognition), remote sensing images exhibit complex structures due to diverse landforms and the distance between acquisition equipment and the Earth—creating a need for accuracy improvement methods. S. Paheding et al. [7] (2024) further noted that deep learning has enhanced the analytical capabilities of remote sensing images, improving classification accuracy and promoting the development of remote sensing imaging platforms and practical applications.

To address accuracy limitations, this paper integrates data augmentation, the CBAM module, GeM pooling, and a small-kernel stem to enhance the model's fitting and generalization abilities. The proposed model is tested on the EuroSAT dataset, and ablation experiments verify the effectiveness of the proposed optimizations. The remainder of this paper is organized as follows: Chapter 2 reviews relevant research; Chapter 3 details the methodology; Chapter 4 presents experimental designs and results; Chapter 5 concludes the study and discusses future directions.

II. Related Work

A. YOLO

Since its initial proposal in 2015, You Only Look Once (YOLO) has undergone multiple major updates. Early YOLOv1 laid the foundation by treating object detection as a single regression task, enabling end-to-end fast detection, but suffered from low accuracy and poor small-object detection performance. Subsequent YOLOv2 improved performance via batch normalization, anchor boxes, and dimension clustering; YOLOv3 further enhanced effectiveness by adopting a more efficient backbone network, multiple anchors, and spatial pyramid pooling.

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In 2020, YOLOv4 introduced innovations such as Mosaic data augmentation, an anchor-free detection head, and an optimized loss function. Recent iterations continue to evolve: YOLOv6 (open-sourced by Meituan in 2022) was applied to autonomous delivery robots, with optimizations focused on model lightweighting and small-object detection; YOLOv8 (released by Ultralytics in 2023) improved performance, flexibility, and efficiency, supporting multiple visual AI tasks. M. Lei et al. [8] (2025) noted that the YOLO series has shifted from prioritizing speed to balancing speed and accuracy, and finally to refined structural design.

YOLO has wide-ranging applications: in autonomous driving, J. Liu et al. [9] (2024) demonstrated its ability to real-time identify vehicles, pedestrians, and traffic signs, providing critical support for safe driving; in video surveillance, H. Zhang et al. [10] (2023) used YOLO for real-time monitoring of abnormal behaviors and object intrusions; in industrial quality inspection, it detects product defects and counts quantities, improving production quality and efficiency.

B. Data Augmentation

1) Data Augmentation in Medical Image Analysis

Z. Xu et al. [11] (2024) proposed an attention-guided cross-domain generative model (CDA-GAN) to generate diverse brain tumor samples. On the BRATS and TCIA datasets, this method increased image classification accuracy and segmentation Dice coefficients by 0.5%–4.18%, effectively alleviating medical data scarcity. C. Tang et al. [12] (2023) integrated data augmentation (random rotation, Gaussian noise injection) with the Inception V3 model, significantly improving the precision and recall of pneumonia classification using chest X-rays and supporting clinical auxiliary diagnosis.

2) Data Augmentation in Computer Vision

F. Pan et al. [13] (2023) proposed the Data-Augmentation PatchmatchNet algorithm, which enhances robustness by adjusting image brightness and removing partial pixels during the input phase, and optimizes information extraction during depth estimation. This approach improved the reconstruction integrity and speed of multi-view stereo vision on datasets such as DTU.

3) Data Augmentation in Speech Recognition

Y. Zhang et al. [14] (2023) addressed the bias of Flemish speech recognition systems toward non-native accents using speed perturbation and SpecAugment; they found that augmenting non-native speech data independently achieved the optimal bias reduction effect, facilitating inclusive speech recognition systems. M. Bartelds et al. [15] (2023) applied data augmentation (self-training for transcribed text generation, Text-to-Speech (TTS) for speech data generation) to low-resource languages (e.g., Gronings), reducing the Word Error Rate (WER) by 20.5%–25.5% and breaking through low-resource speech recognition bottlenecks.

4) Data Augmentation in Natural Language Processing (NLP)

J. Chen et al. [16] (2023) summarized data augmentation methods (word-level, sentence-level, adversarial augmentation) and validated them on 11 datasets (covering classification, reasoning). They found that rational augmentation strategies significantly improve model performance in scenarios with limited annotated data, providing methodological references for practitioners.

III. Methodology

A. Methods of Data Augmentation

To enhance the model's robustness to remote sensing images and reduce overfitting, a multi-dimensional data augmentation strategy was adopted during training:

Random Cropping: A random region (area ratio: [0.8, 1.0] of the original image) is cropped and resized to 224×224 pixels, simulating different viewing angles and forcing the model to focus on key semantic regions.

Random Horizontal Flip: The image is flipped horizontally with 50% probability (no semantic information loss), enhancing adaptability to left-right variations.

Color Jitter: Brightness, contrast, and saturation are perturbed by ±20%, and hue by ±0.2—improving adaptability to different lighting and sensor characteristics while avoiding color distortion.

Random Rotation: The image is rotated by [-20°, +20°], ensuring stability for targets with different orientations.

Tensorization & Normalization: PIL-format images are converted to tensors (pixel values normalized to [0, 1]), then further normalized using the ImageNet dataset's mean (μ = (0.485, 0.456, 0.406)) and standard deviation (σ = (0.229, 0.224, 0.225)) to meet model input requirements and accelerate convergence.

B. Model Structure

An improved deep convolutional neural network based on Residual Network 50 (ResNet50) is proposed, with three key structural optimizations:

1) Small-Kernel Stem: The original 7×7 convolution kernel (stride=2) is replaced with a 3×3 convolution kernel (stride=1), and the first max-pooling layer is removed—reducing spatial resolution loss and enhancing shallow feature extraction.

2) CBAM Integration: A CBAM module is added after each ResNet50 stage. It combines channel attention (focusing on key feature channels) and spatial attention (locating important feature regions) to strengthen the model's focus on effective information in remote sensing images.

3) GeM Pooling: Traditional Global Average Pooling (GAP) is replaced with GeM pooling, which adaptively learns the optimal pooling exponent p based on remote sensing data—enhancing feature discriminability.

The network's classification head maps features to class probabilities:

Flatten Operation: Converts GeM-output feature maps into a one-dimensional vector.

Fully Connected Layer: A 512-dimensional fully connected layer is added, followed by Batch Normalization (BatchNorm), a ReLU activation function, and a Dropout layer (suppressing overfitting).

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Output Layer: Outputs the probability distribution of remote sensing image classes via the Softmax function.

The proposed architecture is shown in Figure 1:

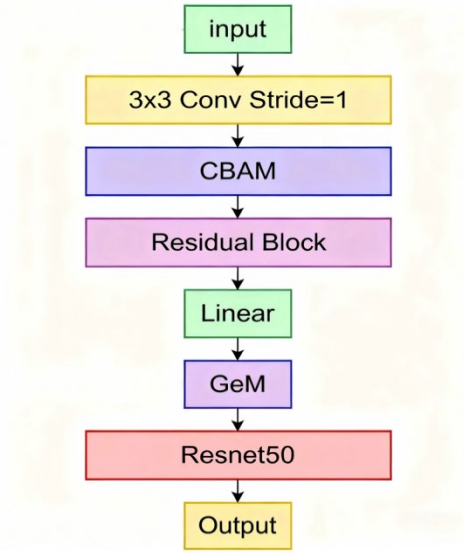


Figure 1. Architecture of the proposed remote sensing image classification model

C. Optimizer and Loss Function

AdamW, a mainstream deep learning optimizer proposed by Loshchilov and Hutter in 2017, features core improvements aimed at addressing the regularization flaws of the classic Adam optimizer. In the Adam optimizer, L2 regularization is implemented by adding a parameter penalty term to gradient calculation. However, Adam's adaptive learning rate (based on second-moment estimation) scales the gradients of different parameters, which "dilutes" the strength of the L2 penalty: specifically, the regularization effect is amplified for parameters with small gradients, while weakened for parameters with large gradients. This ultimately leads to a significant discrepancy between the actual and expected effects of weight decay.

AdamW resolves this issue through a design that "decouples gradient update from weight decay": in each round of parameter update, it first calculates the gradient update amount based on Adam's first moment and second moment to complete the gradient descent operation. Subsequently, it independently performs weight decay on the model parameters (the formula can be expressed as: $\theta_{new} = \theta_{old} \times (1 - \eta \times \lambda)$, where η denotes the learning rate and λ denotes the weight decay coefficient), instead of integrating the penalty term into the gradient. This decoupling mechanism ensures that the strength of weight decay is not affected by the adaptive learning rate, stably constrains the parameter scale, and effectively suppresses overfitting.

In this study, the core reason for selecting the AdamW optimizer lies in its stability in large-scale model training. The current task is based on ResNet50, a medium-to-large visual model. AdamW not only inherits Adam's advantage of "fast convergence speed" but also controls parameter redundancy through stable weight decay. For specific parameter settings: the learning rate is set to 1e-3, which refers to the common initial learning rate of ResNet-series models in image classification tasks. Preliminary small-batch experiments verify that 1e-3 avoids gradient explosion in the early training stage while ensuring rapid model updates; the weight decay coefficient is set to 1e-4, determined through comparative experiments .

Regarding the loss function, we adopt CrossEntropyLoss (cross-entropy loss), the most commonly used loss function in deep learning multi-classification tasks. Its mathematical expression is: $L = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_{ic}\log(p_{ic})$, where N is the number of samples in a batch, C is the number of classes, y_{ic} is the true label of the i-th sample for the c-th class (in one-hot encoding format), and p_{ic} is the predicted probability that the i-th sample belongs to the c-th class (normalized by the Softmax function). The advantages of this loss function are: it can directly measure the difference between the predicted probability distribution and the true label distribution, and its gradient calculation is concise. It is particularly suitable for the current "image category classification" task. Additionally, considering that the dataset has a relatively balanced class distribution, CrossEntropyLoss does not require additional class weights, resulting in higher computational efficiency.

1) AdamW Optimizer

AdamW, proposed by Loshchilov and Hutter (2017), addresses the regularization flaws of the Adam optimizer by decoupling gradient updates from weight decay:

Gradient Update: Calculates the gradient update amount using Adam's first-moment (momentum) and second-moment (adaptive learning rate) estimation.

Weight Decay: Independently applies weight decay to model parameters (Formula (1)): $\theta_{new} = \theta_{old} \times (1 - \eta \times \lambda)$, where η denotes the learning rate and λ denotes the weight decay coefficientParameter Settings: The learning rate is set to 1e-3 , and the weight decay coefficient is 1e-4. AdamW is suitable for ResNet50 (a medium-to-large model) as it inherits Adam's fast convergence while controlling parameter redundancy.

2) CrossEntropyLoss

CrossEntropyLoss is adopted for multi-classification tasks, measuring the difference between predicted probability distributions and true labels (Formula (2)): $L = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_{ic}\log(p_{ic})$, where N is the number of samples in a batch, C is the number of classes, y_{ic} is the true label of the i-th sample for the c-th class (in one-hot encoding format), and p_{ic} is the predicted probability that the i-th sample belongs to the c-th class (normalized by the Softmax function).This loss function is ideal for the study's balanced EuroSAT dataset (no additional class weights required), ensuring concise gradient calculations and efficient training.

3) Training Strategy

Data Preprocessing:

Image Resizing: All original images (192×192 to 512×512 pixels) are resized to 224×224 pixels using bilinear interpolation (balances computational efficiency and detail

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preservation, avoiding jagged distortion from nearest-neighbor interpolation).

Normalization: Uses ImageNet's mean and standard deviation (consistent with ResNet50's pre-training data distribution), reducing the model's cost of adapting to new data and accelerating fine-tuning.

4) Explanation of Data Augmentation Limitations

The study's classification task depends heavily on subtle texture features (e.g., distinguishing land cover types via texture). Aggressive augmentation (e.g., vertical flipping, excessive cropping) can destroy these features, leading the model to learn incorrect patterns and reducing generalization. For future performance improvement (e.g., small-sample scenarios), lightweight augmentation is recommended:

Random horizontal flip (probability=0.5, no vertical flip to avoid structural damage).

Random center cropping (ratio=0.8–1.0, retaining core features).

Fine-tuning of brightness/contrast (±10%, simulating slight lighting changes).

IV. Experiments

A. Experimental Settings

1) Experimental Hyperparameters

Key hyperparameters for model training are shown in Table 1.

TABLE 1: Experimental Hyperparameters

NAME	***VALUE***
IMG_SIZE	224
BATCH_SIZE	64
EPOCHS	30
LR	1e-3
WEIGHT_DECAY	1e-4
USE_PRETRAINED	True
NUM_CLASSES	len(CLASSES)
CHANNEL_REDUCTION	16
GEM_P	3.0
GEM_EPS	1e-6
GEM_LEARN_P	True
CBAM_KERNEL_SIZE	7

B. Dataset Introduction

Supported by the European Space Agency (ESA), the EuroSAT dataset is a satellite remote sensing image classification dataset. Its core purpose is to evaluate the performance of machine learning in remote sensing image scene understanding tasks, and it is widely used in the fields of computer vision and remote sensing image processing.

Built based on high-resolution multispectral remote sensing satellite data from ESA's Sentinel-2, EuroSAT aims to address the scarcity of annotated data in the remote sensing field. It contains 27,000 satellite images covering multiple regions across the European continent, all standardized to a resolution of 64×64 pixels (corresponding to an approximately 32m×32m area on the ground). The dataset provides two data formats:multispectral and RGB, facilitating comparisons with traditional computer vision datasets.

According to land cover types, the dataset is divided into 10 mutually exclusive categories: agricultural land, forest, grassland, highway, industrial areas, residential areas, river, sea/lake, airport, and railway. Each category contains approximately 2,700 images, resulting in a balanced class distribution. The annotations are based on Geographic Information System (GIS) data and expert verification, ensuring high accuracy.

EuroSAT features high authenticity (reflecting the complexity of real remote sensing scenarios), wide versatility (supporting tasks such as image classification, transfer learning, and few-shot learning), and open-source availability. It serves not only as a key tool for intelligent interpretation of remote sensing images and benchmark testing of model performance but also as pre-training data for practical applications such as agricultural growth monitoring and urban expansion analysis.

"Validation accuracy" is defined as the ratio of the number of correctly classified samples to the total number of samples in the validation dataset. This metric evaluates how well a model generalizes to unseen data, providing insight into its performance during training. A higher validation accuracy indicates better generalization ability.

C. Experimental Results

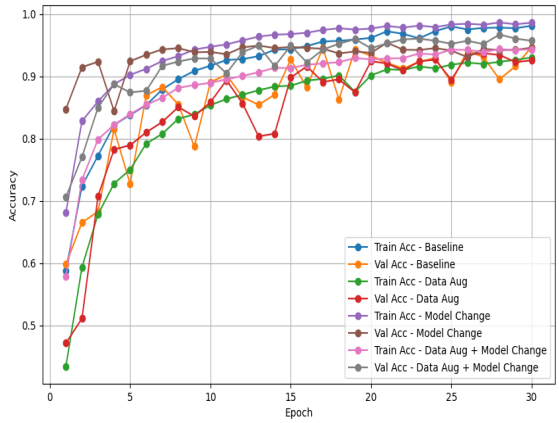


Figure 2. Training and Validation Accuracy Over Epochs

The final experimental results are shown in Figure 2 and Table 2.

TABLE 2: The Final Experimental Results

	baseline	***Data aug***	***Model change***	***Data aug + Model change***
Validation accuracy	0.9474	0.9372	0.9541	0.9672

V. Conclusions

In key fields such as the scientific layout of territorial spatial planning, dynamic monitoring of agricultural production, comprehensive assessment of ecological environments, and

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early warning of natural disasters, the classification and recognition technology of remote sensing images plays an indispensable supporting role, and its application value has attracted much attention. Currently, although some studies have adopted neural network technology to conduct remote sensing image classification work, from the perspective of practical application effects, there is still room for further optimization and improvement in the classification accuracy of these methods, making it difficult to fully meet the demand for high-precision data in various fields.

To address this situation, this paper proposes a brand-new optimization scheme: on the one hand, it expands the diversity of training samples through data augmentation technology; on the other hand, it innovatively integrates and applies the CBAM attention module, GeM pooling operation, and small-kernel stem structure. Starting from multiple dimensions such as the effectiveness of feature extraction, the rationality of feature aggregation, and the scientificity of network initialization, the scheme comprehensively enhances the model's ability to fit the features of complex remote sensing images, while improving the model's generalization performance in different scenarios.

To verify the feasibility and superiority of this scheme, the research team conducted systematic testing on the proposed model using the EuroSAT dataset. The experimental results show that the classification accuracy of the model on the validation set reaches 96.72%, which achieves a significant performance improvement compared with the traditional baseline model (with an accuracy of 94.57%). In addition, this paper also specially designed and completed ablation experiments. By verifying the effects of data augmentation measures and various model improvement modules one by one, it clearly confirms the effectiveness of these optimization methods in improving the classification accuracy of remote sensing images, providing a reliable reference basis for subsequent related research.

However, during the experiment, we found that there is still room for improvement in the accuracy of the experimental results; at the same time, the experimental code runs relatively slowly, and there is a certain room for optimization to improve the code running speed.

REFERENCES

[1] Helber, P., Bischke, B., Dengel, A., & Borth, D. (2019). EuroSAT: A novel dataset and deep learning benchmark for land use and land cover classification. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 12(7), 2217–2226.

[2] Xu, P., Tsendbazar, N. E., Herold, M., De Bruin, S., Koopmans, M., Birch, T., ... & Zanaga, D. (2024). Comparative validation of recent 10 m-resolution global land cover maps. Remote Sensing of Environment, 311, 114316.

[3] Mehmood, M., Shahzad, A., Zafar, B., Shabbir, A., & Ali, N. (2022). Remote sensing image classification: A comprehensive review and applications. Mathematical problems in engineering, 2022(1), 5880959.

[4] Chen, K., Chen, B., Liu, C., Li, W., Zou, Z., & Shi, Z. (2024). Rsmamba: Remote sensing image classification with state space model. IEEE Geoscience and Remote Sensing Letters, 21, 1-5.

[5] Shao, Z., & Cai, J. (2018). Remote sensing image fusion with deep convolutional neural network. IEEE journal of selected topics in applied earth observations and remote sensing, 11(5), 1656-1669.

[6] Kumari, M., & Kaul, A. (2023). Deep learning techniques for remote sensing image scene classification: A comprehensive review, current challenges, and future directions. Concurrency and Computation: Practice and Experience, 35(22), e7733.

[7] Paheding, S., Saleem, A., Siddiqui, M. F. H., Rawashdeh, N., Essa, A., & Reyes, A. A. (2024). Advancing horizons in remote sensing: a comprehensive survey of deep learning models and applications in image classification and beyond. Neural Computing and Applications, 36(27), 16727-16767.

[8]]Lei, M., Li, S., Wu, Y., Hu, H., Zhou, Y., Zheng, X., ... & Gao, Y. (2025). YOLOv13: Real-Time Object Detection with Hypergraph-Enhanced Adaptive Visual Perception. arXiv preprint arXiv:2506.17733.

[9] Cheng, Q., & Chen, L. (2024, August). International Conference on Computer Vision, Robotics, and Automation Engineering (CRAE 2024). In Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series (Vol. 13249).

[10] Zhang, H., Zhou, L., & Lei, J. (2023). IEEE 3rd International Conference on Electronic Technology, Communication and Information (ICETCI).

[11] Xu, Z., Tang, J., Qi, C., Yao, D., Liu, C., Zhan, Y., & Lukasiewicz, T. (2024). Cross-domain attention-guided generative data augmentation for medical image analysis with limited data. Computers in biology and medicine, 168, 107744.

[12] Tang, C., Zhang, T., & Toe, T. T. (2023, December). Data Augmentation Based Inception V3 for Pneumonia Image Classification. In 2023 6th International Conference on Software Engineering and Computer Science (CSECS) (pp. 1-5). IEEE.

[13] Pan, F., Wang, P., Wang, L., & Li, L. (2023). Multi-View Stereo Vision Patchmatch Algorithm Based on Data Augmentation. Sensors, 23(5), 2729.

[14] Zhang, Y., Herygers, A., Patel, T., Yue, Z., & Scharenborg, O. (2023, December). Exploring data augmentation in bias mitigation against non-native-accented speech. In 2023 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU) (pp. 1-8). IEEE.

[15] Bartelds, M., San, N., McDonnell, B., Jurafsky, D., & Wieling, M. (2023). Making more of little data: Improving low-resource automatic speech recognition using data augmentation. arXiv preprint arXiv:2305.10951.

[16] Chen, J., Tam, D., Raffel, C., Bansal, M., & Yang, D. (2023). An empirical survey of data augmentation for limited data learning in nlp. Transactions of the Association for Computational Linguistics, 11, 191-211.

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